

Joint manuscript review packet

Regularized Spectral Statistics for Prime Races and Low-Zero Transient Reversals

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Purpose. This packet contains the complete joint revision, the numerical and formal sources needed to inspect its finite claims, and a precise record of what remains conjectural. The manuscript is ready for joint review; it is not yet cleared for an arXiv replacement or journal submission.

What is included

- submission-shaped TeX and an eight-page compiled PDF;
- the low-zero reconstruction figure, source curve through 3×10^{14} , complete tabular outputs, and executable verifiers;
- the independent modulus-19 Arb certificate and 100-zero stability run;
- compact Lean sources for the corrected character selector and four finite quadratic-nonresidue statements;
- a technical change log, verification report, source diff, provenance, and packet-wide SHA-256 checksums.

Scientific status at a glance

Item	Status
Conjugated character selector	PROVED FINITE ALGEBRA
567-cell baseline comparison	PASS: 567/567
Low-zero reconstruction	VERIFIED FINITE COMPUTATION
Very low odd-character mode mod 19	ARB EXISTENCE CERTIFICATE
Two-parameter regularized limit	CONJECTURE / OPEN
Universal -1 apex and 1 nadir	CONJECTURE / OPEN
Eventual ordinary-count ordering	NOT CLAIMED
Finite- x mollified comparison plot	PENDING DEFINITION

Main numerical findings

The ordinary-count file has 438 checkpoints for $N \in \{7, 8, 11, 19, 23\}$ and agrees exactly with the independently checked baseline in all 567 shared cells through 1.3×10^{13} . The continuation to 3×10^{14} has now been replicated by an independent primesieve-based implementation on Kaggle: 4896/4896 shared cells match exactly at all 72 frontier grid points, and the range counts reproduce the published values of $\pi(10^{14})$, $\pi(2 \times 10^{14})$, and $\pi(3 \times 10^{14})$ exactly (see `numerics/frontier_replication/` in the computational package).

Using 25 positive zeros per nonprincipal character, the top-decade correlations for the normalized -1 trajectories are 0.965, 0.931, 0.939, 0.971, and 0.826. At the final checkpoint, the ranks of -1 are $1/3$,

3/3, 3/5, 6/9, and 1/11, respectively. These continuing reversals motivate regularization but do not test the conjectural hierarchy.

For the odd character of Conrey index 13 modulo 19, an independent Arb sign-change computation certifies a critical-line zero near $\gamma = 0.018956399080226143$. With 100 positive ordinates per nonprincipal character modulo 19, the -1 correlation is 0.9925, while rank and leader changes remain active. The earlier $e^{33.4}$ settling estimate is therefore not retained.

Why the requested mollified plot is pending

The current analytic argument does not yet fix a unique finite statistic, normalization, logarithm branch, or admissible regime $T = T(x)$. It also lacks a summed off-diagonal estimate. A comparison against $C_{N,T} B_N(a)$ before those choices are fixed could be rescaled after the fact. The revision therefore states the two-parameter limit honestly as a conjectural target and supplies the verified ordinary-count reconstruction instead.

Decisions requested from both authors

1. Approve or revise the theorem/conjecture boundary and analytic proof obligations.
2. Confirm title, author order, affiliations, and contribution wording.
3. Confirm the raw-count table and numerical interpretation.
4. Choose a permanent public data/code archive and insert its DOI.
5. Approve one identical final compiled PDF before any arXiv replacement or journal submission.

Start with the top-level README and the PDF in the manuscript folder. Packet file integrity is recorded in the checksum list in the support folder.